VOIDSCOPE · DESIGN DECISIONS REGISTER

VoidScope design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	VDS-DEC-001	Version	0.4
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/06-design-decisions.md @ 463eaf8		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: open decision 2 (concept choice, proposed, awaiting Amish, VDS-DDR-005); STEM fixes F1 and F2 recorded as decided under the standing instruction (VDS-DDR-006); items 15 and 16 to confirm
0.3	2026-10-03	Amish Chadha	Open decision 1 (bite valve access) decided, round 2, 3A (VDS-DDR-004); one item added to confirm when parts are bought; no open decisions
0.2	2026-10-03	Amish Chadha	Amish's requirement decision 3A (steerable camera tip, R11 restated) recorded as made (VDS-DDR-003); one new open decision on bite valve access; four items to confirm when parts are bought; value engineering updated
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; every decision made under Amish's 2026-10-03 pre-approval

VoidScope design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

> **Safety:** Several decisions below set safety stops: the drip ceiling, the bite valve, the blower's shut-off pressure and the battery chemistry. The open decision on bite valve access touches how a survivor is given water. They were taken on the conservative side (VDS-DDR-001, Table 2). VoidScope is rescue equipment that is not certified and is not a medical device.

Open decisions

Open decision 1 (bite valve access) was decided by Amish on 2026-10-03: the valve stays on the nose and the tip is steered aside to give water; see Decisions made and VDS-DDR-004.

#	Decision needed	Options	Recommendation	Affects in the build	Source
2	Concept choice. Proposed, awaiting Amish	A: rigid sectioned rod with steerable tip (design of record); B: reel-fed glass-fibre push rod; C: tripod and winch sonde	A (165 of 205 against 135 and 128; 22 per cent clear of B)	Nothing if A is kept; PRC, PRS, CAL, build plan and BOM redone for B or C	VDS-OPT-001; VDS-DDR-005

- **To confirm when parts are bought**

#	What to confirm	Why it matters	Source
1	The camera head is 29 mm or less across and about 55 mm long, IP68 at 1 m for 30 min, AHD 1080p with about a 90° view and 12 dimmable LEDs, with a pigtail at least 170 mm long	It sets the tip housing bore, R3 and R4; the pigtail must reach back through the tip to its splice in the nose	VDS-DDR-001 D4; VDS-CAL-001 C
2	The blower's shut-off (maximum static) pressure is 1.0 kPa or less, and its curve gives at least 20 L/min through 0.38 kPa	It is the air pressure safety stop (R6)	VDS-DDR-001 D8, S3; VDS-CAL-001 E
3	The flow meter's pressure drop is 0.2 kPa or less at 20 L/min and it fits 100 mm tall	A larger drop lowers the full-speed flow; the lid must close	VDS-CAL-001 E; VDS-DDR-002 P11
4	The snap button suits 1.47 mm and 2.11 mm walls, its button stands about 1.5 mm proud, and it holds a 300 N pull	It locks every joint	VDS-DDR-002 P2; VDS-CAL-001 B
5	The two tube sizes telescope with about 0.11 mm a side as bought	The couplers must slide in by hand	VDS-DDR-002 P1
6	The capillary bore is 0.5 mm, 750 mm long (was 600 mm), and the measured flow at 1 m and 20 °C is about 1.2 mL/min	It sets the drip ceiling (R5)	VDS-DDR-001 S1; VDS-CAL-001 D
7	The bite valve is about 11 mm across, fits the 4 mm tube and stays shut under 20 kPa of water head	It is the water safety stop	VDS-DDR-001 S2; VDS-CAL-001 D7
8	The speaker is 20 mm across and 4 mm thick, IP67	It must sit flush in its pocket	VDS-DDR-002 P8
9	The hybrid cable is 6 mm or less across, and the case, monitor (28 mm deep) and battery (151 x 65 x 94 mm) match the sizes modelled	They set the grommets, the plate layout and the lid stack	VDS-DDR-002 P10, P11
10	The 0.6 mm 7x7 stainless wire rope breaks at 250 N or more	It sets the factor of 3.0 at a 30 N thumb push against the stop	VDS-DDR-003; VDS-CAL-001 J16
11	The 3 mm steering housings bend to about 15 mm radius where they turn up into the control body, and their friction is about 0.15	A stiffer housing needs a taller pad; more friction raises the thumb force	VDS-DDR-003 T5, T8; VDS-CAL-001 J13
12	The camera cable's bending stiffness is near 5,000 N·mm²	It sets the thumb force to hold 90° (about 5 N)	VDS-CAL-001 J12 to J14
13	The printed links' 2.0 mm pin holes ream cleanly and the lugs hold the pins without rocking	The joints must swing freely but not wobble the camera	VDS-DDR-003 T3
14	Confirm in drills with the medical lead that a survivor can reach the bite valve on the nose past the tip steered fully aside	The valve stays on the nose (decision 3A, round 2); if it is hard to reach, a 60 mm flexible valve extension is the fallback	VDS-DDR-004
15	The camera's electrical power at the dimmed LED setting is about 0.8 W or less, and the black anodising of the tip	It sets the tip surface at 44 C at 35 C ambient (PR-014)	VDS-DDR-006 F1; VDS-CAL-001 K3

#	What to confirm	Why it matters	Source
	housing is available and quoted (it is not priced in the BOM)		
16	The suppliers' data for the camera (power, dimmer), blower (shut-off pressure) and bite valve (cracking pressure) are in hand before any build	They are the three figures the safety stops rest on	VDS-STM-001 S11, S14, S18

Value engineering

Value-engineering target: USD 2,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,780 (USD 220 under the target). Main cost drivers and savings worth trying:

- The largest lines are the lathe time allowance (USD 240), the monitor with recorder and bracket (USD 175), the cleaning kit and four spare water sets (USD 118), the rod sections (USD 128 for four), the rod carry case (USD 115), the battery packs (USD 110 for two), the surface unit case (USD 110) and the camera head (USD 95).
- Savings worth trying: a builder with a lathe saves USD 240; a soft carry bag in place of the hard rod case saves about USD 80; a single battery pack saves USD 55 but leaves no spare for a full shift at -10 °C; printing the base link and drum in PA12 with the links would save about an hour of lathe time (USD 60) at some cost in wear.
- Making the design constructable added the couplers, buttons and pins, the guide tube, the equipment plate, the brackets and the lathe allowance, about USD 290 in all. The steerable tip (decision 3A) added USD 202: links, pins and sheaths USD 42, base link USD 6, tip housing USD 8, wires and housings USD 38, steering control USD 48 and one more hour of lathe time USD 60.

Decisions made

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review items D1 to D13: lines inside with the bore as air duct; 38.1 mm aluminium sections with snap buttons; no articulation; bought IP68 camera head; monitor in the surface unit lid; tip speaker as speaker and microphone; gravity drip with capillary ceiling and bite valve; blower chosen by shut-off pressure; two LiFePO4 packs; independent of the crawler block; requirements restated; budget, pitch and problem kept	Amish: "I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."	VDS-DDR-001
2026-10-03	Conservative safety choices S1 to S6, each with the evidence that would relax it	Amish, same pre-approval	VDS-DDR-001, Table 2
2026-10-03	First co-design candidate to approach: the disaster management training school of the Armed Police Force, Nepal; second: Turkey's AFAD training centres (not contacted, not partners)	Amish, same pre-approval	VDS-DDR-001 D11
2026-10-03	Design for construction, changes P1 to P14	Amish, same pre-approval	VDS-DDR-002
2026-10-03	R11 (two 45° bends) recorded as not met in the first version; a flexible lead section is a later version, not TRL 3 work (superseded the same day by decision 3A below)	Amish, same pre-approval	VDS-REQ-001 v0.3; VDS-DDR-001 D12
2026-10-03	Appearance model departures from `model.py` (rounded edges, depth bands, labels, cables drawn to the surface unit, rubble context) accepted for renders only	Amish, same pre-approval	`docs/REVIEW.md`, TRL 3
2026-10-03	Decision 3A: add a cable-steered articulating camera tip that bends up to 90° each way, controlled from the handle; restate R11 as "see around a 45-degree bend from a straight hole" with a verification method; keep the rigid rod, the air path and the water tube to the bite valve; the tip must still pass the 51 mm hole	Amish: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	VDS-DDR-003; VDS-REQ-001 v0.4
2026-10-03	3A (round 2): bite valve kept on the nose; the drill card says steer the tip aside to give water; "confirm in drills with the medical lead" added under To confirm; articulating camera tips added to the patent screen list. No design change	Amish: "i agree with all the 46 recommendations you provided. please proceed."	VDS-DDR-004
2026-10-24	STEM fix F1: tip housing black anodised and LEDs dimmed by default (heat in the tip, 85 C to 54 C at 45 C ambient); no change of working principle	Standing instruction of 2026-10-23, applied by Claude	VDS-DDR-006
2026-10-24	STEM fixes F3 (capillary 750 mm), F4 (tip touch criterion 48 C for 10 min, 35 C ambient) and F5 (blower from the regulated converter with a speed cap) after the independent review	Standing instruction of 2026-10-23, applied by Claude	VDS-DDR-006
2026-10-24	STEM fix F2: option B reach restated as floor-supported only, free reach capped at 1 m	Standing instruction of 2026-10-23, applied by Claude	VDS-DDR-006
2026-10-24	Concept choice: option A recommended, Proposed, awaiting Amish (not decided)	Proposed, awaiting Amish	VDS-DDR-005